

Depth

§1. Structure theory

§2. Representation theory

na local field

$$F \supset \underset{\substack{\uparrow \\ \mathfrak{o}}}{\mathfrak{o}} \longrightarrow \mathfrak{o}/(\varpi) = k \text{ residue field, char } k = p,$$

$$\text{val} : F \rightarrow \mathbb{Z} \cup \{\infty\}$$

$$\varpi \mapsto 1$$

(more generally, any discretely valued henselian local field w perfect res. field)

§1.1 $G(F)^\circ$

Goal: G conn'd reductive F -group $\rightsquigarrow G(F)^\circ \subset G(F)$

when G simply conn'd, $G(F)^\circ = G(F)$

Step 1. T F -torus $\rightsquigarrow$ lft Néron model $T / \mathfrak{o}$

$\cup$

connected N. model T° conn'd special fibre

$$T(F)^\circ := T^\circ(\mathfrak{o})$$

Step 2. G quasi-split $/F$, Choose Borel $B \supset T$ max'l F -torus

$$G^{sc} \rightarrow DG \text{ simply conn'd cover}$$

$$G(F)^\natural := \text{im}(G^{sc}(F) \rightarrow G(F))$$

$$G(F)^\circ := G(F)^\natural T(F)^\circ$$

Step 3. More general situation. e.g. G anisotropic, but then Bruhat-Tits theory is trivial



§1.2 Bruhat-Tits buildings

G conn'd reductive F -group $\xrightarrow{BT}$ polysimplicial set $B(G)$ (building)

w/ properties:

- ① $B(G)$ is equipped w/ a $G(F)$ -action
- ② $B(G)$ has a collection of subcomplexes $\{A\}$ called apartments

Apartments $\xleftrightarrow{\text{bijection}} \text{max'l split } k\text{-tori of } G$

$$A(S) \longleftrightarrow S$$

$G(F)$ -equivariant: $g \cdot A(S) = A(gSg^{-1})$

- ③ $A(S)$ is an affine space under $X_*(S)_{\mathbb{R}}$

The simplicial str. on $A(S)$ is given by the subdivision determined by

- ④ $B(G)$ satisfies tamely ramified descent. affine coroots
- eg. $G = SL_2$, $F = \mathbb{Q}_p$, infinite graph, each vertex has valence $p+1$.



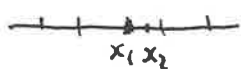
every line extends to ∞ is an apartment



Def. A parahoric subgroup of $G(F)$ is a group of the form

$$G_x := \text{Stab}_{G(F)}(x) \cap G(F)^{\circ}, \quad x \in B(G)$$

eg.



$$G_{x_1} = SL_2(\mathbb{O}) \text{ (hyperspecial)}$$

$$G_{x_2} = \left(\begin{smallmatrix} \mathbb{O} & \mathbb{O} \\ \varpi \mathbb{O} & \mathbb{O} \end{smallmatrix} \right)_{\det=1} \text{ (Iwahori)}$$

Thm (Bruhat-Tits) $\exists$ smooth affine $\mathbb{O}$ -model g_x of G , st. $g_x(\mathbb{O}) = G_x$.

Σ

§1.3 Moy-Prasad filtration Goal: decreasing filtration $G_x \supset G_{x,r}$, $r \in \mathbb{R}_{\geq 0}$.

G split reductive gp. Fix a split max'l torus T. Up to $G(F)$ -conjugation,

can take $x \in A(T) \subset B(G)$.

Fix a Chevalley system $\{x_\alpha: G_\alpha \xrightarrow{\sim} U_\alpha\}_{\alpha \in \Phi(G,T)}$:

$\forall w \in \text{Weyl group} = N_G(T)(F)/T(F)$, $\exists$ lift $\tilde{w}$ s.t.

$$\tilde{w} x_\alpha(c) \tilde{w}^{-1} = x_{w(\alpha)}(\pm c).$$

Filt'n of torus. $T(F)_0 = \{t \in T(F): \text{val}(x(t)) \geq 0, \forall x \in X^*(T)\} (= T(F)^\circ)$

$$r > 0, T(F)_r = \{t \in T(F)_0: \text{val}(x(t)-1) \geq r, \forall x \in X^*(T)\}$$

Filt'n of U_α . $x \in A(T)$ ^{fixed chamber} $\xrightarrow{\sim} X_*(T)_{\mathbb{R}}$

$$r \geq 0, U_\alpha(F)_{x,r} := x_\alpha(\omega^{[r - \langle \alpha, x \rangle]} \mathcal{O})$$

Filt'n of G_x . $r \geq 0, G_{x,r} := \langle T(F)_r, U_\alpha(F)_{x,r} : \alpha \in \Phi(G,T) \rangle$

eg. $G = SL_2$.

$$\begin{array}{c} | \quad | \quad | \quad | \\ x_1 \quad x_2 \end{array}$$

$$G_{x_1,0} = SL_2(\mathcal{O}), \quad G_{x_1,r} = \begin{pmatrix} 1 + \omega^{[r]} \mathcal{O} & \omega^{[r]} \mathcal{O} \\ \omega^{[r]} \mathcal{O} & 1 + \omega^{[r]} \mathcal{O} \end{pmatrix}_{\det=1}, \quad r > 0$$

$$G_{x_2,0} = \begin{pmatrix} \mathcal{O} & \mathcal{O} \\ \omega \mathcal{O} & \mathcal{O} \end{pmatrix}_{\det=1}, \quad G_{x_2,r} = \begin{pmatrix} 1 + \omega^{[r]} \mathcal{O} & \omega^{[r-\frac{1}{2}]} \mathcal{O} \\ \omega^{[r+\frac{1}{2}]} \mathcal{O} & 1 + \omega^{[r]} \mathcal{O} \end{pmatrix}_{\det=1}, \quad r > 0$$

Filt'n on $\mathfrak{g} \otimes \mathcal{O}^*$

$$\mathfrak{t}_r = \{x \in \mathfrak{t}(F): \text{val}(d\chi(x)) \geq r, \forall \chi \in X^*(T)\}$$

$$\mathfrak{g}_{\alpha,x,r} := \omega^{[r - \langle \alpha, x \rangle]} \mathcal{O} \otimes x_\alpha(1) \subset \mathfrak{g}_\alpha, \quad \alpha \in \Phi(G,T)$$

$$\mathfrak{g}_{x,r} := \mathfrak{t} \oplus \bigoplus_{\alpha \in \Phi(G,T)} \mathfrak{g}_{\alpha,x,r}.$$

$$\mathfrak{g}_{x,-r}^* := \{x \in \mathfrak{g}^* : X(Y) \in \omega \mathcal{O}, \forall Y \in \mathfrak{g}_{x,s}, s > r\}$$

Tamely ramified case. G/F splits after a tame ext'n E/F .

$$r > 0, \quad G(F)_{x,r} := G(E)_{x,r}^{G(E|F)}, \quad g, g^* \dots$$

have an explicit description using relative root systems.

Properties

$$① \quad G_{x,0} = G_x, \quad G_{x,r} \text{ open cpt}, \quad \bigcap_r G_{x,r} = \{e\}$$

$$G_{x,r+} := \bigcup_{s>r} G_{x,s}$$

r is a jump at x if $G_{x,r} \neq G_{x,r+}$.

$$J(x) := \text{jump at } x \text{ is discrete,} \quad J(x)^* := J(x) - \{0\}.$$

$$② \quad [G_{x,r}, G_{x,s}] \subset G_{x,r+s}$$

$\leadsto G_{x,r}/G_{x,r+}$ abelian for $r > 0$

in fact, has nat'l str. of a k -vec. sp.

$$③ \quad r > 0, \quad G_{x,r} \text{ is pro-} p.$$

$$④ \quad g \in G(F), \quad g G_{x,r} g^{-1} = G_{g \cdot x, r}, \quad g, g^* \dots$$

$$⑤ \quad (g_x)_k \twoheadrightarrow \bar{G}_x \text{ reductive quotient,} \quad G_x/G_{x,0+} \simeq \bar{G}_x(k).$$

$$⑥ \quad G_{x,r} \supset G_{y,s} \quad (r, s > 0) \Rightarrow g_{x,r} \supset g_{y,s}$$

$$⑦ \quad J(x)^* = \{r_1 < r_2 < \dots\}, \quad \exists \text{ isom. of } k\text{-vec. sp.}$$

$$G_{x,r_i}/G_{x,r_{i+1}} \xrightarrow{\sim} g_{x,r_i}/g_{x,r_{i+1}}, \text{ cpt'ble w } G_x/G_{x,0+}\text{-action}$$

$$\exists \text{ non-deg. } k\text{-bilinear form } \langle, \rangle : g_{x,-r_i}^* / g_{x,-r_{i+1}}^* \times g_{x,r_i}/g_{x,r_{i+1}} \rightarrow k$$

that is $G_x/G_{x,0+}$ - equiv.

§1.4 Stratum

Def A stratum of depth $\begin{cases} r > 0 \\ 0 \end{cases}$ is a coset space of $\mathfrak{g}_{x, -r}^* / \mathfrak{g}_{x, -r+}^*$, some $x \in \mathfrak{g}_{x, 0+}^*$, some x

$X \in \mathfrak{g}^*$ is nilp. $\iff \exists$ cocharacter $\lambda: \mathbb{G}_m \rightarrow G$ s.t. $\lim_{t \rightarrow 0} \text{ad}^* \lambda(t) \cdot X = 0$.

A stratum $\mathfrak{X}$ of depth $r > 0$ is minimal if $\mathfrak{X}$ doesn't contain a nilp. elt.

Any stratum of depth 0 is declared to be minimal.

Prop. $\mathfrak{X}$ minimal stratum of depth $r > 0$, $\mathfrak{X} \cap \mathfrak{g}_{y, -s}^* = \emptyset$, $\forall s < r$, $y \in \mathcal{B}(G)$

Def Two strata are associated if their G -orbits have nonempty intersection.

Prop 1) $\mathfrak{X}$ minimal stratum of depth r , any stratum $\mathfrak{Y}$ assoc. w/ $\mathfrak{X}$ has depth $\geq r$.

2) Two associated minimal strata have the same level.

§1.5 Optimal points

h_x : only depends on the facet $F \ni x$

$h_{x,r}$ ($r > 0$): depends on x , not just F . But many points give essentially

equiv. filters (problem of linear programming)

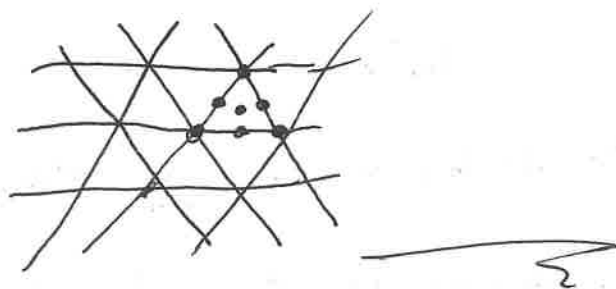
Slogan. Up to $G(F)$ -conjugation, only finitely many points need to be considered.

Thm. $\exists$ finite set $P_0 \subset \mathcal{B}(G)$ (optimal points) s.t.

① the jumps at an optimal pt are rat'l numbers. They form a countable set, denoted by $I(G)$.

② $J(x)^* = \{r_1 < r_2 < \dots\}$, $x \in g_{x, -r_i}^* / g_{x, -r_{i-1}}^*$ stratum contains a nilp. elt then $\exists y \in \mathfrak{p}_0$, $g \in \mathfrak{g}(F)$, $s \in J(y)$ s.t. $x \in g_{s, y, -s}^*$.

eg. SL_3



R noetherian $\mathbb{Z}[\frac{1}{p}]$ -comm. algebra.

$$\text{Rep}_R(G) = \text{cat. of smooth } R\text{-reps of } G(F) \supset \text{Irr}_R(G)$$

§ 2.1 Unrefined types

Def An unrefined type of depth $\begin{cases} r > 0 & \text{is a character } g_{x, r} / g_{x, r+1} \rightarrow R^\times \\ 0 & \text{is a trivial char. } g_{x, 0} \rightarrow R^\times. \end{cases}$

Fix a nontrivial additive char. $\psi: k \rightarrow R^\times$

$$J(x)^* = \{r_1 < r_2 < \dots\}$$

Unrefined types $\chi: g_{x, r_i} / g_{x, r_{i+1}} \rightarrow R^\times \iff$ stratum $x = x + g_{x, -r_{i-1}}^* \in g_{x, -r_i}^*$
 $\exists$ is $g_{x, r_i} / g_{x, r_{i+1}}$

$$\chi(g) = \psi(\langle x, \Xi(g) \rangle)$$

χ minimal if the corresponding x is minimal.

triv: $g_{x, 0} \rightarrow R^\times \iff$ depth 0 stratum $g_{x, 0}^*$

it is declared to be minimal

Prop. Any $0 \neq V \in \text{Rep}_R(G)$, $\exists$ nonzero map $X \rightarrow V$, X minimal unrefined type assoc. to $X \in \mathcal{PO}$ of depth $r \in I(G)$

Proof

Let r be the smallest elt of $I(G)$ s.t. $\exists X \in \mathcal{PO}$, $\bigvee_{W \in \mathcal{W}} G_{X,r+} \neq 0$.

$r=0$, done

$r > 1$. $G_{X,r}$ pro-p, $W \in \text{Rep}_R(G_{X,r})$ is semisimple.

If $\exists$ non-minimal unrefined type of $G_{X,r}/G_{X,r+}$ contained in W ,
 $\Rightarrow \bigvee G_{g,y,s+} \neq 0$ for some $y \in \mathcal{PO}$, $g \in G(F)$, $s \in I(G)$, $s < r$.

Contradicts to the minimality of r . $\square$



§ 2.2 Depth of rep's

$\pi \in \text{Irr}_R(G)$

Prop-Def. ① The minimal unrefined types that map nontrivially to π have the same depth, called the depth of π , denoted by $r(\pi)$

② $r(\pi) = \text{smallest } r \geq 0 \text{ s.t. } \pi \bigwedge_{X \in \mathcal{B}(G)} G_{X,r+} \neq 0 \text{ for some } X \in \mathcal{B}(G), r \in J(X)$.

Proof. ① The inputs:

Lemma 1. K, K' open cpt subgroups of $G(F)$, $\pi \in \text{Rep}_R(G)$ generated by $RG(F)$

-mod. by K -submodule $\sigma \in \pi|_K$. Every irred. quotient σ' of $\pi|_{K'}$

is a quotient of $c\text{-ind}_{g(K) \cap K'}^{K'} = \text{res}_{g(K) \cap K'}^{g(K)}(g\sigma)$ for some $g \in G(F)$.

Proof. Consequence of Mackey formula.

Lemma 2. μ : unrified type of depth $s > 0 \iff \gamma = Y + g_{Y, -s+}^* \in g_{Y, -s}^*$
 χ :  depth $r > 0 \iff \chi = X + g_{X, -r+}^* \in g_{X, -r}^*$

then $\chi = \mu$ on $G_{X,r} \cap G_{Y,s}$ if and only if $\chi \cap \gamma \neq \emptyset$.

↓

χ, γ are associated

↓ χ, γ minimal

$\text{depth}(\chi) = \text{depth}(\gamma)$

① follows from Lemm 1 & 2.

②: $\pi|_{G_{X,r+}} \neq 0 \Rightarrow \exists$ unrified type of depth r maps to π

$\Rightarrow r \geq r(\pi)$

Prop. The set of depths of minimal unrified types $= I(G)$.

Pr. $(G_{X,r}, \chi)$ minimal unrified type of depth $r > 0$

$P_\chi := \text{c-ind}_{G_{X,r}}^{G(F)} \chi$ is gen'd by χ .

$\exists$ minimal unrified type of depth $s \in I(G)$ maps to P_χ

By similar argument, $r = s \Rightarrow r \in I(G)$.

Prop-Def. A rep'n $V \in \text{Rep}_R(G)$ is said to have depth r if all minimal unrified types map to it are of depth r .

In fact, V is of depth $r \iff V$ is gen. by minimal unrified types of depth r .



§ 2.3 Depth decomposition

$\text{Rep}_R(G)_r :=$ subcat. of depth r representations, R -linear abelian cat.

Goal. $\exists$ decomposition of R -linear abelian cats $\text{Rep}_R(G) \simeq \prod_{r \in I(G)} \text{Rep}_R(G)_r$.

Decomposing abelian cat.

$\mathcal{C}$ (R -linear) abelian cat. $\{P(r)\}$ a family of objects

[PROJ] Each $P(r)$ is compact projective.

[DISJ] $r \neq s, \text{Hom}_{\mathcal{C}}(P(r), P(s)) = 0$

[GEN] $\forall 0 \neq V \in \mathcal{C}, \text{Hom}_{\mathcal{C}}(\bigoplus_r P(r), V) \neq 0$

Then $\mathcal{C} \simeq \prod_r \mathcal{C}_r$, where

$$\mathcal{C}_r = \{V \in \mathcal{C} : \text{Hom}_{\mathcal{C}}(P(s), V) = 0, \forall s \neq r\}$$

In our situation, $\mathcal{C} = \text{Rep}_R(G)$,

$$r > 0, P(r) = \bigoplus_{\substack{x \in P_0, \\ r \in J(x)}} c\text{-ind}_{G_{x,r}}^{G(F)} x$$

over $x \in P_0, r \in J(x), (G_{x,r}, x)$ minimal unrefined type
finite direct sum.

$$P(0) = \bigoplus_{x \in P_0} c\text{-ind}_{G_{x,0}}^{G(F)} \text{triv}$$

$P(r)$ compact projective: $c\text{-ind}$ has an exact right adjoint.

[DISJ], [GEN] follow from previous discussion $\Rightarrow$ desired decomposition.



§2.4. Application. TFAE.

(1) $\forall$ open cpt subgroup H of $G(F)$, the Hecke algebra $R[H \backslash G(F) / H]$

is f.g. over its centre, and its centre is a f.g. R -alg.

(2) Any finitely gen. $R[G]$ module V is $\mathfrak{f}$ -finite: $\mathfrak{f}$ - image of $\mathfrak{f} \rightarrow \text{End}_{R[G]}(V)$ is f.g. R -algebra
 $\uparrow$
 Bernstein centre - V is admissible over $\mathfrak{f}$.

Pt. (2) $\Rightarrow$ (1) $V = \text{c-ind}_H^{G(F)}(R) = R[G(F)/H]$ f.g.

$$\text{End}_{R[G(F)]}(V) \cong R[H \backslash G(F)/H]$$

the action of $\mathfrak{f}$ on V is given by a morphism $\mathfrak{f} \rightarrow Z(R[H \backslash G(F)/H])$
 image $=: \mathfrak{f}_V$

By assumption, V is $\mathfrak{f}$ -finite

$\Rightarrow V^H = R[H \backslash G(F)/H]$ is f.g. as $\mathfrak{f}_V$ -module

$\sim$ f.g. as $Z(R[H \backslash G(F)/H])$ -mod

By def'n, $\mathfrak{f}_V$ f.g. R -alg $\Rightarrow$ noetherian $\Rightarrow Z(R[H \backslash G(F)/H])$ f.g. $\mathfrak{f}_V$ -mod
 $\Rightarrow$ f.g. as R -alg.

(1) $\Rightarrow$ (2). H open pro- p subgroup of G , $R[G(F)/H] \cong \bigoplus_r R[G(F)/H]_r$.

$\sim R[H \backslash G(F)/H] = \text{End}_{R[G(F)]}(R[G(F)/H])$, $Z(R[H \backslash G(F)/H])$ decompose
 Fix r . accordingly

$\text{Rep}_R(G)_r$ is gen. by a cpt projective generator.

$\rightarrow$ can find H s.t. $R[G/H]_r$ is a compact projective generator
 for $\text{Rep}_R(G)_r$.

Morita equiv: $\text{Rep}_R(G/H)_r \cong \text{Mod-End}_{R[G]}(R[G/H]_r)$

$\Rightarrow \mathfrak{f}_r = Z(\text{Rep}_R(G/H)_r) = Z(\text{End}_{R[G]}(R[G/H]_r)) = Z(R[H \backslash G/H]_r)$

$\Rightarrow R[G/H]_r$ is $\mathfrak{f}$ -finite. True for all $r \Rightarrow$ any fin. gen.

V is $\mathfrak{f}$ -finite.